Detector Design Spec 1 — 4H-SiC Dosimetry Diode (p-n, zero-bias capable)

For: foundry hand-off (ST / FBK / other) From: 4H-SiC detector TCAD feasibility study Date: 2026-06-16 Status: process-agnostic physics specification (NOT a mask set)

> How to read this document. This is a physics target the foundry maps to its own process — not a mask layout. We do not have ST/FBK design rules, achievable doping windows, or epi grids, so every "fab fills in" cell below is deliberately left to the foundry. The electrostatic targets (doping profile, depletion, full-depletion voltage, CCE-when-depleted) are TCAD model outputs. The 2D-vs-1D-twin C-V agreement is a mesh/implementation consistency check, not a validation: the 2D mesh is laterally uniform, so 2D = 1D by construction (they agree to ~1e-14). See §6 for what the C-V fit does and does not show, including the one independent external check we have. The leakage, breakdown, and zero-bias-efficiency rows are explicitly NOT predictive — see the caveats.

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1. Device concept

Planar one-sided p^+/n^- junction on a graded n-type epitaxial layer over an n^+ (or p^+-substrate, per the telescope convention) wafer, with a guard ring for edge termination. Intended to operate under reverse bias for clinical dosimetry, and to retain a self-powered (zero-bias) mode via the built-in junction field.

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2. Electrostatic target spec (TCAD model targets + tolerance — fab maps to process)

Parameter	Target value	Tolerance (fab maps)	Basis
Junction type	p^+/n^-, one-sided	—	design choice
Epi thickness (biased clinical device)	10 µm	±1 µm	nominal; epi/n^+ position set by the C-V fit
Graded epi donor profile N_D(y)	N_D(y) = N_D_bulk + (N_D_junction − N_D_bulk)·exp(−y/L)	see below	fit to digitised measured C-V (Photons Fig. 4b)
→ N_D_junction (near junction)	8.19×10^{15} cm^{-3}	±20 %	C-V fit (not resolved inside W(0 V))
→ N_D_bulk (deep epi)	3.38×10^{13} cm^{-3}	±20 %	C-V fit; below the documented $0.5–1 \times 10^{14}$
→ L (grading length)	0.46 µm	±0.3 µm	C-V fit
Graded n^-/n^+ transition	last 1.14 µm of the epi, log-linear to 4.5×10^{16} cm^{-3} at 10 µm	fab: SIMS / SRP	C-V fit (−10…−60 V creep), not SIMS
n^+ substrate (cathode side)	10^{19} cm^{-3} beyond the epi end (1 µm modelled)	—	device stack (owner-confirmed)
p^+ anode doping N_A (total)	1×10^{19} cm^{-3}	fab implant/anneal	active ≈ 1.3×10^{18} cm^{-3} (≈13 % ionized, model)
Junction depth (p^+ emitter)	~0.3 µm	0.3-1 µm (no effect on W)	Petringa 2025 Fig. 1
Relative permittivity ε_r	9.7	—	4H-SiC literature
Guard ring	required	geometry per fab design rules	edge termination

Depletion behaviour (W = ε·A/C of the small-signal C, as the measured W). The whole digitised measured curve (every 250 mV, 0 to −60 V) is a FIT INPUT, not a TCAD prediction: plateau rms 0.013 µm over −5…−59 V, knee rms 0.24 µm (−3.25…−0.25 V; the exponential epi form cannot follow the measured dip to ~1.3×10^{13} at 4–6 µm). The n/N_D = 0.5 depletion edge is 1.65 / 9.48 / 9.74 µm:

Reverse bias	W = ε·A/C (model)	measured	Note
0 V	1.655 µm	1.68	built-in depletion (self-powered drift region)
−2 V	6.94 µm	7.18	knee
−5 V	9.25 µm	9.27	low-doped epi depleted (V_fd 2.7 V, below)
−10 V	9.52 µm	9.52	edge creeping into the graded n^-/n^+ transition
−30 V	9.75 µm	9.73	idem (in-fit residual −0.02 µm)

What the fit does and does not fix. The profile reproduces the measured C-V; it is not a measured doping profile. N_D_bulk 3.4 × 10^{13} is below every documented statement for this epi (0.5–1 × 10^{14}; the authors' "order 10^{14}" is the 1/C² slope over the knee window, −6…−2 V: 1.2 × 10^{14}). The position of the epi/n^+ interface and the shape of the transition are set by the fit, not by SIMS: C-V does not show where 10^{19} is reached. SIMS/SRP through the epi/n^+ interface, or C-V to −100…−200 V (model: ε·A/C ≈ 9.96–9.98 µm at −100 V, above 10 µm at −200 V), would settle it. The previous default (`etna.nplus_calibrated_config()`, abrupt epi/n^+, calibrated on 0 / −10 V) put the knee at −8…−10 V and missed −30 V by +2.6 %; it is excluded by the full curve (lab/cv_gap/notes/LOG.md).

The model is the p^+ / graded n^- epi / n^+ substrate stack: the cathode sits on the n^+, and devsim converges to ≥ −150 V (1D) / −100 V (2D, the parity-checked range). The old ~−50 V "solver wall" existed only in the legacy model (ohmic cathode directly on the n^- epi, `etna.legacy_config()`); it was a contact artefact, never punch-through or breakdown. There is still no rated voltage and no breakdown model — see §5.

Full-depletion voltage V_fd (analytic, graded profile, integrated to the START of the graded n^-/n^+ transition): 2.74 V (10 µm epi); ~0.7 V (5 µm) and ~11.5 V (20 µm) with the same 1.14 µm transition carried to those thicknesses (an extrapolation: the fit is on the 10 µm device). The drift-diffusion reach-through (`reach_through_V`, edge within one Debye length of the transition start) is 3.10 V for the 10 µm epi: the two definitions agree. Depleting the transition itself is the −10…−60 V creep of the measured C-V, not full depletion of the sensitive epi. (Previous default, abrupt epi/n^+: V_fd 9.93 V, reach-through 12.3 V; legacy profile: 10.4 V.)

Operating bias. The validity gate used by the sweep code is ≥ 1.2 × V_fd = 3.3 V (the 20 % margin covers ±20 % doping: V_fd 3.2 V at N_D_bulk + 20 %, 3.9 V with both N_D + 20 %). The paper's −10 V is 3× that gate (3.6× V_fd), −30 V is 9× (11× V_fd). Recommendation: keep ≥ −10 V as the practical operating point. The model CCE is already 0.998 at −3 V, but −10 V costs nothing in leakage terms (3.99 pA model estimate on 4 mm²), is where the measured C-V is flat (W within 0.4 µm of its −60 V value), and covers what the C-V fit cannot pin: if the bulk is at the documented 0.5–1 × 10^{14} instead of the fitted 3.4 × 10^{13}, V_fd rises to 3.9–7.6 V, still below −10 V with margin. Use −30 V where the extra margin is wanted (it is the condition of the published measurements).

Charge collection (reverse bias, TCAD-model, consistent with the 1D twin): CCE → 1 (> 0.96) once the device is fully depleted. This is a drift-diffusion model result for the simulated p-n junction; not an absolute device guarantee. CCE is relative to the e-h pairs generated inside the simulated epi, not to the full particle deposit: for the model alpha (5.486 MeV, NIST ASTAR Bragg curve in SiC, 18.4 µm range, deposit peaking near 17 µm; since 2026-10-20) a 10 µm epi holds ~43 % of the pairs (5 µm: ~20 %, 20 µm: ~100 %), so collected/deposited charge ≈ 0.43 × CCE on 10 µm at normal incidence (the earlier 15 µm flat-box model gave ~83 % / ~42 % / ~100 %; the 2D lateral scans report both, as `in_mesh_generation_fraction` and `q_collected_over_deposited`).

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3. Zero-bias (self-powered) operation — honest scope

The built-in p^+/n^- junction gives a depletion region W(0 V) = 1.65 µm (ε·A/C; measured 1.68 µm) with a built-in field that separates carriers without applied bias — so the device is electrostatically self-powered. However, on the 10 µm clinical epi only ~17 % of the epi sits in the 0 V drift field; charge generated deeper collects by diffusion, which our model does not quantify at 0 V. So:

- Defensible now: the device is self-powered; the drift-active 0 V volume is ~1.7 µm. (The measured knee is steep: W ≈ 3.9 µm at −1 V, 7.2 µm at −2 V, so a small bias already depletes most of the epi.)

- NOT claimed: a quantitative 0 V collection efficiency on the 10 µm device — that needs transient DD modeling plus one measured zero-bias anchor.

Design lever for a genuinely zero-bias dosimeter (recommended separate variant): $W(0\ V) \propto \sqrt{(V_{bi} \cdot \varepsilon / (q \cdot N_D))}$. To get a _meaningful depleted fraction at 0 V, use a thinner, lower-doped epi: target ~3–4 µm epi at $N_D \approx 1–3 \times 10^{14}$ cm⁻³, which depletes ≥ 80 % of the epi at 0 V ($W(0\ V) \approx 3–6$ µm). This trades signal depth for self-powered operation. The fitted 8.2×10^{15} junction profile above is the biased clinical device — NOT the zero-bias variant. If a dedicated zero-bias detector is wanted, we will re-run the calibration at the thin/low-doped target and quote its model W(0 V).

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4. Fab fills in (explicitly not modeled here)

- As-grown epi profile, exact peak doping, doping tolerances → fab process window.
- p^+ implant species/dose/anneal and actual activation (we model ~13 % ionization).
- Contact metallization stack, anneal, ohmic contact resistivity, sheet resistance.
- Passivation, surface preparation.
- Guard-ring geometry and edge-termination implementation per design rules.
- Achievable epi-thickness grid and minimum feature sizes.

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5. Caveats — do NOT read these as committed specs

- Dark current / leakage: ESTIMATE ONLY. $N_t = 1.356 \times 10^{12}$ cm⁻³ s⁻¹ ($E_t = 0.67$ eV) is fitted so that the model equals the experimental upper bound "< 18 pA up to −60 V" at −60 V, 4 mm², 300 K (18.000 pA) — a calibration input, not a prediction or validation. The bias shape (3.99 / 11.88 / 18.00 pA at −10 / −30 / −60 V) comes from a heuristic depletion weight min(E / 3.5 × 10⁴ V/cm, 1) times the Hurkx (1992) trap-assisted-tunnelling enhancement (1 + Γ(E)) (active since 2026-10-20: Γ ≈ 1.8 at 10⁵ V/cm, 300 K, with an assumed tunnelling mass 0.25 m_0; it replaced a branch that kept Γ = 1 at all fields here and gave 6.3 / 15.8 / 18.0 pA); the "TAT" term is an effective field-weighted generation; the SRH/TAT/SRV split is not physically meaningful; and the temperature dependence $N_t(T) \propto n_i(T)$ is an assumption (~×15 per +13 K), not fitted. (Before 2026-10-16 a stale-field bug made the model look consistent with 18 pA at −30 V; on the corrected path the unscaled default gave ~141 pA, i.e. 7.8× over the bound.) Use only as a budget estimate; the fab must measure. Do not treat it as a leakage limit/guarantee.
- No rated breakdown voltage. There is no impact-ionization/breakdown model. (The ~−50 V convergence limit of earlier versions belonged to the legacy model with the cathode on the n^- epi; it was a solver artefact, not a rated or physical limit.)
- Edge/guard-ring performance not modeled — our mesh is a laterally-uniform 1D-twin; it cannot predict edge field crowding or termination breakdown.
- CCE figures are DD-model results for the simulated junction, checked for consistency with the 1D twin only.

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6. What is solid (model targets, within the limits above)

Electrostatics: built-in potential, depletion vs bias, electric field, C-V, the graded doping profile fitted on the measured C-V, full-depletion voltage, and CCE-vs-bias when depleted (relative to in-epi generation). These are the dimensioning core and are defensible as model targets with the tolerance bands above. The 2D-vs-1D-twin C-V agreement is a mesh/implementation consistency check (laterally uniform mesh, so 2D = 1D by construction), not an independent validation.

On the C-V agreement: the default profile is FITTED on the digitised measured C-V of this device (Photons Fig. 4b, 230 points; plateau rms 0.013 µm, knee rms 0.24 µm), so its agreement with that curve is a fit quality, not a validation. (The historical $R^2 = 0.998$ figure scored `calibrate_graded_doping()` against the same three points it was fitted on: a fit-recovery check, likewise not a validation.) The only literature comparison available is against Bruzzi & Verroi ("Epitaxial SiC Dosimeters and Flux Monitoring Detectors for Proton Therapy Beams," Materials 2023, 16(10),

3643), an unrelated published 4H-SiC device our model was never calibrated to: ETNA's predicted depletion width (ε·A/C) at −40 V deviates 3.1% from that paper's reported endpoint; that is the Schottky-vs-pn built-in-voltage difference (<1 % with V_bi matched, Schottky on a 30 µm epi at 2.46e15: a different device, NOT a validation of this one). The measured C-V of THIS device (Photons Fig. 4b) puts the knee at −1…−4 V; the previous abrupt-n^+ default put it at −8…−10 V, which is why the default is now the C-V fit (lab/cv_gap). This is a solver-scale sanity check (two fitted endpoints, not a dense point-by-point overlay) — see `.planning/VALIDATION_LITERATURE.md` for the full comparison and caveats. The electrostatics model itself (drift-diffusion/Poisson) is independently checked against Sze & Ng textbook forms and is not in question here — only the $R^2 = 0.998$ validation claim, as originally worded, overstated what it demonstrated.